

WEEK 5 DAY 4 — MEMORY + CONVERSATION INTELLIGENCE

BLOCK 1 — CORE IDEA

Right now your system: User → question → answer 

👉 BUT: Every question is independent  (no memory)

Problem Example

User: What is AVL tree?

Bot: Answer 

User: Why is it better?

Bot:  no context → fails

After Day 4

User: What is AVL tree?

Bot: 

User: Why is it better?

Bot:  understands AVL context

GOAL

👉 Add: Conversation memory

BLOCK 2 — DESIGN

 You already have (GOOD):

```
self.chat_history = deque(...)
```

👉 BUT you are NOT using it properly for reasoning 

✅ New Flow

User Query



Combine:

- Chat history ✅
- Retrieved context ✅



Send to LLM



Better answer ✅

🔥 BLOCK 3 — IMPLEMENTATION (YOUR EXACT CODE FIX)

✅ STEP 1 — Modify `retrive_context()`

👉 Add history awareness:

```
def retrive_context(self, user_text):
```

```
    # ✅ combine last 2 messages for context
```

```
    history_text = " ".join([ m["content"] for m in list(self.chat_history)[-3:] ])
```

```
    full_query = history_text + " " + user_text
```

```
    # ✅ use full_query instead of user_text
```

```
    embedded_user = self.embed(full_query)
```

```
    chroma_chunks = self.loader.top_3_chunks(embedded_user)
```

```
    bm25_chunks = self.loader.bm25.search(full_query, top_k=3)
```

```
    merged = reciprocal_rank_fusion(bm25_chunks, chroma_chunks)
```

```
    top_chunks = merged[:3]
```

```
if not top_chunks:
    return "No Context Found"
return "\n\n".join(top_chunks)
```

✅ STEP 2 — Improve react_chat()

👉 Add chat history explicitly:

Replace: context += f"\nUser Question: {user_input}\n"

✅ WITH:

```
history = "\n".join([f"{m['role']}: {m['content']}" for m in self.chat_history])
context += f"\nConversation History:\n{history}\n"
context += f"\nUser Question: {user_input}\n"
```

✅ Now LLM sees:

user: what is AVL

assistant: ...

user: why better

👉 AND understands it ✅

🔥 STEP 3 — ADD MEMORY LIMIT (IMPORTANT)

✅ Prevent history from growing too large

Update:

```
def trim_history(self):
    while len(self.chat_history) > 10:
        self.chat_history.popleft()
```

Then call after each message: `self.trim_history()`

✅ STEP 4 — TEST CASES (IMPORTANT)

✅ Test 1 (context carry)

User: What is AVL tree?

User: Why is it better?

✅ Now should work

✅ Test 2 (follow-up)

User: Explain heap

User: What is its time complexity?

✅ Should refer to heap

✅ Test 3 (switch topic)

User: What is BST?

User: What about AVL?

✅ Should still work

💡 BLOCK 4 — WHY THIS IS POWERFUL

✅ Before

Stateless bot ❌

✅ After

Conversational AI ✅

✅ This is EXACTLY what ChatGPT does

🔥 BONUS (OPTIONAL — VERY POWERFUL)

✅ Add "last topic memory"

```
self.last_topic = ""
```

Update inside chat:

```
if "AVL" in user_:
```

```
    self.last_topic = "AVL"
```

Then:

```
full_query = self.last_topic + " " + user_text
```

👉 This improves follow-up accuracy

✅ ✅ FINAL CHECKLIST

Make sure:

✅ History passed to retrieval

✅ History passed to LLM

✅ Memory trimmed

✅ Follow-up questions work

🚀 YOUR PROGRESS

Day 1 → BM25 ✅

Day 2 → Hybrid ✅

Day 3 → API ✅

Day 4 → Memory ✅ 🔥
